Data Structures — Final Exam (2023–2024 Academic Year, Semester 1)

> **Sealing line (left margin)**: College: ____ Major: ____ Class: ____ Student ID: ____; I pledge to abide by the examination rules and take this exam honestly, without cheating. Student name (signature): _____; Answers written inside the sealing line will not be graded.

Question	I	II	III	IV	V	VI	Total
Score							

I. Multiple-Choice Questions (2 points each, 20 points total)

1. The time complexity of searching for an element in an array is ()

a. O(1) b. O(log n) c. O(n) d. O(n log n)

2. Which of the following descriptions of a stack is correct ()

a. first-in, first-out b. last-in, first-out c. first-in, last-out d. random access

3. What is the maximum number of nodes on level n of a complete binary tree ()

a. n b. 2^n c. 2^(n-1) d. n^2

4. The time complexity of inserting a node into a linked list is ()

a. O(1) b. O(log n) c. O(n) d. O(n log n)

5. Among the following sorting algorithms, which has the best time complexity ()

a. bubble sort b. insertion sort c. selection sort d. quicksort

6. The average search time complexity of a hash table is ()

a. O(1) b. O(log n) c. O(n) d. O(n log n)

7. Among the storage structures for graphs, the one suited to representing sparse graphs is ()

a. adjacency matrix b. adjacency list c. multiple adjacency matrix d. edge-set array

8. The worst-case time complexity of searching for an element in a binary search tree is ()

a. O(1) b. O(log n) c. O(n) d. O(n log n)

9. In heapsort, the time complexity of building the max-heap is ()

a. O(1) b. O(log n) c. O(n) d. O(n log n)

10. Which of the following algorithms is a typical application of divide and conquer ()

a. linear search b. binary search c. depth-first search d. quicksort

II. Fill-in-the-Blank Questions (2 points each, 20 points total)

1. In a linked list, the time complexity of deleting a node is _____.

2. The hash function in a hash table is mainly used for _____.

3. The depth-first search (DFS) algorithm is usually implemented with the _____ data structure.

4. The breadth-first search (BFS) algorithm is usually implemented with the _____ data structure.

5. The average time complexity of quicksort is _____.

6. In an AVL tree, the difference in height between the left and right subtrees of any node does not exceed _____.

7. The main algorithms for the minimum spanning tree of a graph are Prim's algorithm and the _____ algorithm.

8. Topological sorting is used to process _____ graphs.

9. In the level-order traversal of a binary tree, the _____ data structure is usually used to assist the implementation.

10. Huffman coding is a _____ code.

III. True-or-False Questions (2 points each, 10 points total)

1. A stack is a first-in, first-out (FIFO) data structure. ()

2. A red-black tree is a self-balancing binary search tree. ()

3. In a linked list, the time complexity of searching for an element is O(log n). ()

4. In a hash table, open addressing is one of the methods for resolving collisions. ()

5. Dijkstra's algorithm can be used to solve the shortest-path problem in unweighted graphs. ()

IV. Short-Answer Questions (5 points each, 20 points total)

1. Briefly describe the main differences between stacks and queues, and give typical application scenarios for each.

2. Describe the inorder traversal process of a binary search tree and explain its property.

3. Explain what a hash collision is and the common methods for resolving it.

4. Explain the working principles of depth-first search (DFS) and breadth-first search (BFS) on graphs.

V. Comprehensive Questions (10 points each, 20 points total)

1. Given an unordered array, design an algorithm to sort it, and analyze the algorithm's time complexity and space complexity.

2. For shortest-path algorithms on graphs, explain the basic steps of Dijkstra's algorithm and demonstrate it with an example.

VI. Programming Question (10 points total)

1. Write a program that implements the basic operations of a linked list (insertion, deletion, search) and tests their functionality. It must be written in C or Java, and the code must include appropriate comments.